

Team Name	COWGIRLS	
Member1	Yoojin Jang	Food Science, Product Manager
Member2	Arim Lee	Coumputer Science, Developer

❑ 1. PROJECT OVERVIEW

1.1 Project titlez

Smart Farm Solution for Rural-Urban Synergy via Wind-Customized Ammonia Scrubbing

1.2 Key challenge areas

Urban Air Quality and Climate Resilience

1.3 Solution

Integrated Solution (Hardware + Software)

1.4 Project Summary

Ammonia gas from livestock farms can travel with the wind and contribute to fine dust pollution in urban areas. However, large-scale rural livestock farms often have little incentive to manage it due to high chemical costs and limited local complaints.

To address this issue, we propose smart farm software that integrates weather API data with an algorithm for estimating ammonia emissions from livestock. When the estimated emission level exceeds a predefined threshold, the software automatically activates selected fence valves facing urban areas to spray neutralizing agents. This converts ammonia into ammonium salt fertilizer, reducing chemical costs, supporting eco-friendly subsidy compliance, and improving urban air quality through urban-rural synergy.

❑ 2. CONTEXT AND PROBLEM STATEMENT

2.1 Environmental Context and Urgency

Large-scale livestock farms emit ammonia gas from manure every day, and this gas can travel with the wind to urban areas, contributing to the formation of fine particulate matter. However, rural farms have little incentive to continuously operate existing odor-reduction equipment due to limited local complaints and high neutralizer costs.

2.2 Scale

¹⁾Approximately **8,500** large-scale beef cattle farms (owning over 100 heads) and **5,400** dairy farms nationwide.

2.3 Stakeholders

Wholesale Livestock Farms (Producers): Located in blind spots with no complaints; want government subsidies but abandon systems due to high operational (chemical) costs.

Urban Residents (Victims): Primary victims facing respiratory health risks from rural-born fine dust.

Government and Local Authorities (Regulators): Waste budgets by distributing hardware without software management, failing to reduce actual urban PM2.5 levels.

2.4 Negative Impacts and Consequences

Environmental and Health Damage: Increased fine dust alerts and massive social medical costs for citizens.

Policy Failure: Expensive environmental infrastructure is left unused, wasting national budgets and deepening rural-urban environmental conflicts.

❑ 3. EXISTING SOLUTIONS AND GAP ANALYSIS

3.1 Currently Available Solutions in the Market

- **Large-scale Biofilters Systems:** Hardware systems designed to capture and filter ammonia gas inside the farm.

1) Statistics Korea. (2026). Number of Households and Head of Livestock by Province and Rearing Scale (Hanwoo, Beef Cattle) Available at: <https://kosis.kr/>

- **Manual or Timer-based Misting Systems:** Spraying devices that operate manually or on a simple timed schedule to spread neutralizing agents.

3.2 Limitations and Gap Analysis

- **Financial Barriers:** Biofilters require thousands of dollars for setup/maintenance, making adoption impossible for independent wholesale farms.
- **Chemical and Resource Waste:** Timer systems spray 24/7 regardless of weather, wasting expensive chemicals when there is no wind or when wind blows away from target zones.
- **The Missing Gap:** Lack of an **intelligent software** that minimizes operational costs through real-time, wind-based precision control

■ 4. PROPOSED SOLUTION AND CORE FEATURES

Our solution is a **Wind-Customized Ammonia Mitigation Software** that integrates real-time weather APIs with a livestock ammonia emission estimation algorithm. When estimated emissions exceed a threshold, the software automatically activates specific fence valves in the direction of urban areas to spray neutralizing agents, preventing pollution at a minimal cost.

Core Features

1. **Ammonia Calculation** (JAES Formula): Computes real-time livestock ammonia production by applying the empirical estimation formula from the Journal of Animal Environmental Science (JAES, 2022).
2. **Smart Spray & Direction Control:** Dynamically regulates both the reasonable amount and valve direction of the neutralizing spray by processing real-time wind speed, wind direction, and emission levels simultaneously.
3. **Eco-Subsidy Data Logging:** Tracks operation minutes and chemical cost savings, generating a visualized data report for farmers to easily apply for government eco-friendly subsidies.

■ 5. INNOVATION & COMPETITIVE ADVANTAGE

5.1 Innovation

- **Chemical Transformation:** Moves beyond simple odor masking. It combines real-time weather data with chemical scrubbing to transform airborne ammonia into solid ammonium salt—a high-quality fertilizer component—safely dropping it to the ground.

5.2 Competitive Advantage

- **Low-Cost Software Solution:** Eliminates the need for expensive hardware upgrades. Instead of purchasing new machinery, farmers simply install smart valves and our software onto their existing misting systems, drastically reducing initial investment.

5.3 Feasibility & Development Plan

- We will build an IoT prototype using a scaled-down farm testbed that simulates real-time weather-driven valve control across directional fence zones. The system architecture is designed to be directly integrated into existing misting infrastructure in large-scale livestock farms.

■ 6. TARGET GROUPS & POTENTIAL IMPACT

6.1 Target Audience

- **Wholesale (B2B) Livestock Farms Outside Cities:** Farms in regulatory blind spots that lack motivation for emission control but need to reduce chemical costs and receive government eco-subsidies.

6.2 Potential Impact

- **Environmental:** Drastically reduces urban fine dust (PM2.5) by blocking ammonia on its pathway and promotes resource recycling through gas-to-fertilizer transformation.
- **Economic:** Maximizes farm profits by saving up to 70% on chemical costs while securing government subsidies; increases government budget efficiency.
- **Social:** Lowers respiratory medical costs for urban residents and resolves environmental conflicts between rural and urban areas.

■ 7. DESCRIPTION OF TECHNOLOGIES APPLIED

7.1 Proposed Technologies

- **Mobile Frontend:** Android Native
- **IoT:** C++(Arduino IDE)
- **Backend Server:** FastAPI
- **Protocol:** HTTPS REST API & MQTT

7.2 System Architecture

The system adopts a 4-Layer Automated Control Architecture (Data → Control → Edge → Actuator) to block urban-bound ammonia gas with minimal resource consumption. The core software logic consists of three sequential pipeline components:

① Dynamic Emission Estimation Model

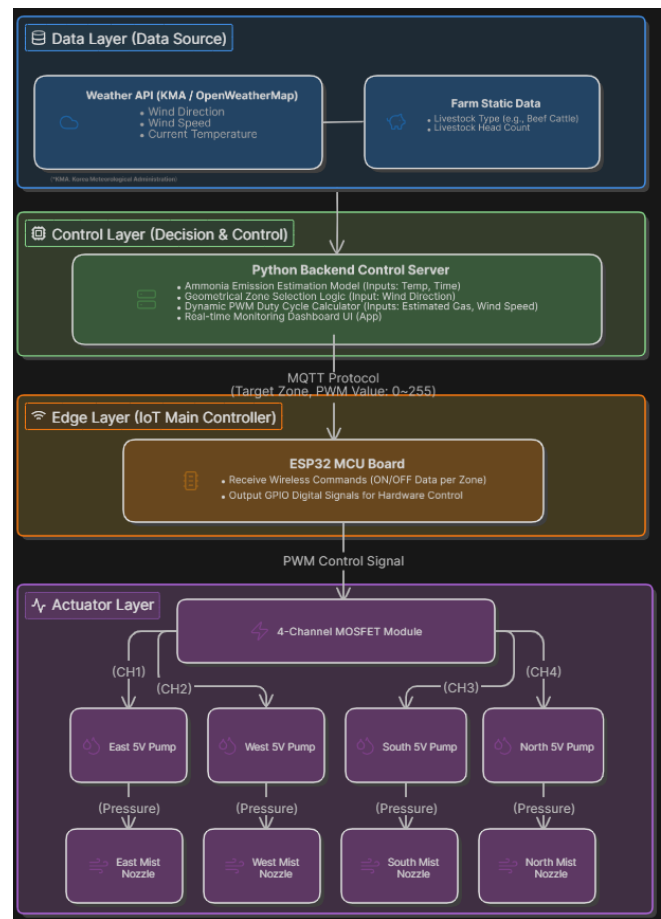
Based on established research data from the National Institute of Animal Science (NIAS), this model estimates the real-time total ammonia emission of the livestock house by factoring in variables of current temperature and current time.

② Geometrical Zone Selection Logic

It analyzes the collected real-time wind direction azimuth (0° to 360°) to determine the target blocking sectors (zone activation flags among East, West, South, and North), which represent the path where the gas disperses toward the city.

③ Dynamic PWM Duty Cycle Calculator

Intersects the estimated gas volume with real-time wind speed to determine the optimal spraying pressure (PWM: 0-255), transmitting the control packet via MQTT to the ESP32.



7.3 Data and Infrastructure Requirements

• Data

- OpenWeatherMap API
- Farm Static Data

• Hardware Requirements

- 1 x ESP32 MCU Board
- 1 x 4-Channel MOSFET Module
- 4 x 5V DC Submersible Pumps & Fine Mist Nozzle Kits

8. REFERENCES

-Park, S., Hwang, O., Yang, B., & Park, J. (2022). A study on ammonia emissions from mechanically-ventilated weaned pig house. *Journal of Animal Environmental Science*, 24(2), 86-90. <https://doi.org/10.11109/JAES.2022.24.2.086>

- Ministry of Agriculture, Food and Rural Affairs (MAFRA), "2024 Guidelines for Livestock Odor Mitigation Project": Support for Misting Systems and Odor Reduction Infrastructure [In Korean]

- Rural Development Administration. (2019). *Reduction measures for fine dust and ammonia in the agricultural and livestock sectors by 2022* (Joint Press Release). [In Korean] https://www.rda.go.kr/board/board.do?prgId=day_farmprmninfoEntry&dataNo=100000756326